De-Americanization

Removing the early-exercise premium from listed option quotes, fast

Vol-Fitter Technical Notes, No. 05

Vol-Fitter

June 28, 2026

Abstract

Listed single-stock and ETF options are *American*: they may be exercised early, so their prices carry an early-exercise premium over the European value. Feeding those prices straight into a European implied-volatility inversion biases the smile — upward, and worst where vega is small. This note documents the Vol-Fitter's de-Americanization: for each quote it finds the volatility σ^* such that a Cox–Ross–Rubinstein American tree reprices the quote, and returns σ^* as the European-equivalent Black volatility. We explain why this tree inversion is the right control variate, how discrete cash dividends enter through an escrowed-spot lattice, and the numerical choices — tree depth, the bracketed-bisection sweep — that hold the inversion to sub-vol-bp accuracy. The hot path is a Numba kernel that de-Americanizes a whole chain with no per-step allocations: on a 300-quote chain it runs in 0.03 ms versus 11.5 ms for the NumPy batch, a 427× speed-up, and a content-digest cache ensures fit-tuning never re-runs it. A worked example shows the naive European inversion over-stating implied volatility by up to 282 volatility basis points on a dividend-paying name, an error the de-Americanization removes entirely.

Contents

1 Why de-Americanize **3**

2 The pricing tree **3**

2.1 Cox–Ross–Rubinstein 3

2.2 Discrete cash dividends 4

3 The de-Americanization map **4**

3.1 Tree inversion as a control variate 4

3.2 Static bounds and bracketing 4

4 Numerical choices **5**

4.1 Depth and bisection 5

4.2 The Numba kernel 5

4.3 The content-digest cache 5

5 Worked example and measured cost **5**

6 What is original, and future directions **6**

A Hyperparameter atlas **6**

B Performance notes **7**

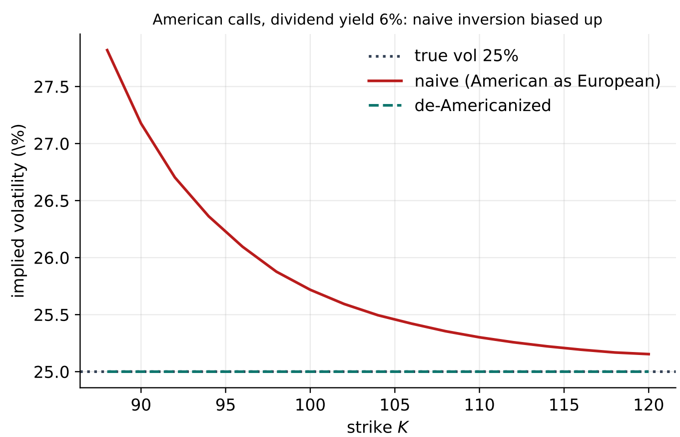
1 Why de-Americanize

The volatility models of Notes 01–04 all assume *European* prices: they work with $C_T(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+]$, a terminal expectation. A listed American option is worth more, because its holder may exercise before expiry; the excess is the *early-exercise premium* (EEP). If one inverts an American price with the European Black formula, the extra value is misattributed to volatility, and the implied vol comes out biased.

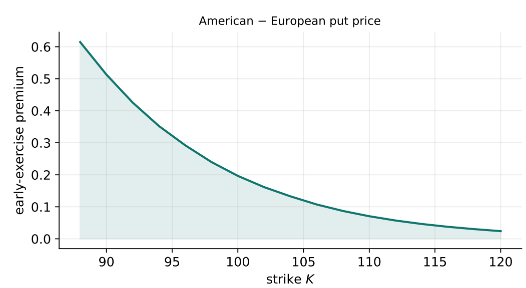
Heuristic

The bias is not uniform. It is largest exactly where the European model is least sensitive to volatility — deep in-the-money, where vega is small, so a given dollar of early-exercise premium translates into a large spurious vol move. A naive smile built from American quotes therefore has corrupted wings, the very region the fitter most needs to extrapolate. De-Americanization is not a cosmetic preprocessing step; it is the difference between fitting the market's volatility and fitting its exercise policy.

Figure 1 makes this concrete on American calls written on a stock paying a 6% dividend yield (where early exercise to capture dividends is rational): an option priced at a true 25% volatility, inverted naively as European, implies a volatility up to 282 vol bps too high, rising as the option goes in the money. De-Americanization recovers a flat 25%.



(a) naive vs de-Americanized implied vol



(b) the early-exercise premium

Figure 1: The naive European inversion of an American price is biased up by the early-exercise premium; the de-Americanization returns the true volatility.

2 The pricing tree

2.1 Cox–Ross–Rubinstein

The American value is computed by a Cox–Ross–Rubinstein (CRR) binomial tree. With N steps of size $\Delta t = T/N$, the spot moves by $u = e^{\sigma\sqrt{\Delta t}}$ or $d = 1/u$ each step, with risk-neutral up-probability

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \qquad 0 < p < 1, \tag{1}$$

(r the rate, q a continuous dividend yield). The condition $0 < p < 1$ is the CRR no-arbitrage requirement $d < e^{(r-q)\Delta t} < u$; it fails only for extreme drift-to-vol ratios and then *raises* rather than silently mis-pricing. The tree rolls back from the payoff with discounting $e^{-r\Delta t}$, taking $\max(\text{continuation}, \text{intrinsic})$ at each node for the American feature. The European value is the same recursion without the intrinsic comparison.

2.2 Discrete cash dividends

A continuous yield q is a crude proxy for a real single name. Discrete cash dividends are handled by the *escrowed-spot* method: the present value of dividends with ex-dates in $(0, T]$ is stripped from the spot to form the diffusing "base" lattice, and the actual spot at each node is the base lattice plus the present value of dividends not yet paid. This keeps the lattice recombining (cash dividends otherwise break recombination) while pricing early exercise against the *actual* cum-dividend spot. The two mechanisms coexist: discrete cash for near dividends, a continuous proxy for the far tail.

3 The de-Americanization map

3.1 Tree inversion as a control variate

De-Americanization is a one-dimensional root find. Given an American quote A at (s, K, T, r, q),

$$\boxed{\text{find } \sigma^* : \ \mathrm{CRR}_{\mathrm{Am}}(\sigma^*) = A, \quad \text{return } \sigma^* \text{ as the European-equivalent Black vol.}} \tag{2}$$

Heuristic

Why is the *tree's* σ^* the right European volatility? Because the tree is used as a *control variate* for the early-exercise feature. The quantity we ultimately want is the European Black vol; the market price differs from the European price by an EEP we cannot observe. The tree supplies a model EEP as a function of σ. Solving $\mathrm{CRR}_{\mathrm{Am}}(\sigma^*) = A$ implicitly subtracts the model EEP at σ^*, and the European Black price at the *same* σ^* equals the tree's European value (the two agree to tree discretization, $\sim 10^{-5}$ at the default depth). So σ^* is exactly the de-Americanized Black vol: the early-exercise premium cancels in the control variate, leaving the diffusion volatility. The method is exact in the model and robust in practice because the EEP is a smooth, slowly varying function of σ.

3.2 Static bounds and bracketing

A price below intrinsic (impossible for an American option) or above the static upper bound (s for a call, K for a put) yields `nan`, mirroring the fitter's convention for unusable quotes. The bracket lower end is lifted just above the CRR drift floor $\sigma\sqrt{\Delta t} > |r - q|\Delta t$ so (1) stays valid, and the upper end is expanded until the price is bracketed, capped at $\sigma = \texttt{SIGMA\_HI} = 4$.

4 Numerical choices

4.1 Depth and bisection

Two depths are used. The *scalar* path (one quote at a time) uses $N = \texttt{DEFAULT\_STEPS} = 501$ and a Brent root find; the European leg of the tree then matches Black to $\sim 10^{-5}$. The *batch* path uses a coarser $N = 192$ tree and a fixed 24-sweep bracketed bisection, which prices within a few 10^{-4} of Black and resolves σ^* to ~ 0.002 vol bps — far below the fit budget. The bisection count was tuned down from 45 to 24 at no measurable accuracy cost (< 0.01 vol bp drift), a $\sim 2\times$ saving on the isolated de-Am rail.

Caution

The tree depth is a numerical-target knob, not a free speed dial. Cutting the batch depth from 192 to 128 buys a $\sim 4\times$ speed-up but shifts the priced European leg by ~ 15 vol bps — i.e. it changes the *answer*, not just the runtime. The depth is held at 192 and speed is bought elsewhere (the Numba kernel, section 4.2).

4.2 The Numba kernel

The hot path de-Americanizes a whole chain. The NumPy batch vectorizes across quotes but still allocates per step; the compiled kernel does a single scalar CRR per quote with no per-step allocation — precomputed power tables, the escrow lattice constants hoisted out, and a `prange` parallel loop over quotes.

Performance

On a 300-quote chain the Numba kernel runs in 0.03 ms versus 11.5 ms for the NumPy batch — a $427\times$ speed-up (the documented figure on wide real chains is $\sim 60\times$; this synthetic chain, being uniform, parallelizes especially well). If Numba is unavailable the code falls back to the NumPy batch with identical results to bisection rounding. The kernel agrees with the scalar tree to tree-rounding precision.

4.3 The content-digest cache

De-Americanization depends only on the de-Am *inputs* — forward, discount, the cash-dividend schedule, the year fractions, and the as-of — not on any fit setting. The pipeline keys a cache on a content digest of exactly those inputs, so tuning a fit penalty, changing the model, or re-weighting quotes never re-runs the trees. The de-Am cost is paid once per genuine data change and is then free across the entire calibration sweep. A conservative pre-screen (non-positive bids, a wing buffer) drops quotes that cannot survive the static bounds *before* the tree runs, and is output-preserving by construction.

5 Worked example and measured cost

Figure 1 is generated by pricing American calls on a 6%-dividend stock at a true 25% volatility and inverting each both ways. The naive European inversion over-states the volatility by up to 282 vol bps, rising into the money exactly as section 1 predicts; the de-Americanization returns a flat 25%. The early-exercise premium (panel b) is the economic source of the bias: positive, and largest in the money where early exercise to capture the dividend is most valuable. The timing table (table 1) records the kernel speed-up on a wide chain.

Table 1: De-Americanizing a 300-quote chain (fastest of three repeats, 192-step trees, 24 bisections).

Path	time (ms)	relative
NumPy batch	11.5	1×
Numba kernel	0.03	427×

6 What is original, and future directions

The control-variate view of tree inversion is folklore; the contributions here are the *allocation-free Numba kernel* (the per-step allocation, not the arithmetic, was the NumPy bottleneck), the *content-digest cache* that scopes de-Am cost to genuine data changes rather than to fit iterations, and the *measured* depth/bisection tuning that cut work without moving the numerical target. The natural next step — documented but not yet shipped — is to seed the bracket with a fast analytic American approximation (Barone–Adesi–Whaley or Bjerksund–Stensland 2002) so each tree starts near its root, composing with the kernel for a further constant-factor win; these are *seeds*, not the answer, so they change no contract.

Caution

De-Americanization inherits the model's assumptions: a flat dividend treatment, no discrete jumps, and the CRR diffusion. For names with hard-to-model dividend timing or borrow, σ^* is only as good as the forward and dividend inputs (Note 06). The method removes the *early-exercise* bias; it cannot repair a wrong forward.

A Hyperparameter atlas

Table 2: De-Americanization hyperparameters (all hidden / internal).

Constant	Default	Role
`DEFAULT_STEPS`	501	Scalar CRR tree depth; European leg matches Black to $\sim 10^{-5}$.
`DEFAULT_BATCH_STEPS`	192	Batch/kernel tree depth.
`BATCH_BISECTIONS`	24	Bracketed-bisection sweeps (~ 0.002 vol bp).
`SIGMA_LO`	10^{-4}	Lower vol bracket (lifted above the CRR drift floor).
`SIGMA_HI`	4.0	Upper vol bracket cap.
escrow PV	—	Discrete-cash-dividend escrowed-spot lattice.
pre-screen buffer	1.5×	Wing buffer for the conservative bid/bounds pre-screen.

Constant	Default	Role
content digest	—	Cache key (forward, discount, dividends, t, τ, as-of).

B Performance notes

1. **Numba kernel** (section 4.2). Single scalar CRR per quote, no per-step allocation, precomputed powers, `prange` over quotes; measured 427× over the NumPy batch on a 300-quote chain. Graceful NumPy fallback.
2. **Bisection tuning.** $45 \to 24$ sweeps, < 0.01 vol bp drift, $\sim 2\times$ on the isolated rail.
3. **Content-digest cache.** De-Am re-runs only on a genuine data change, never on fit-setting changes — so it is free across a calibration sweep.
4. **Output-preserving pre-screen.** Drops unusable quotes before the tree, by construction not changing any surviving result.
5. **Future (not shipped):** BAW / BS-2002 analytic bracket seeds to start each tree near its root.

References

[1] J. Cox, S. Ross and M. Rubinstein. Option pricing: a simplified approach. *J. Financial Economics*, 7(3):229–263, 1979.

[2] G. Barone-Adesi and R. Whaley. Efficient analytic approximation of American option values. *J. Finance*, 42(2):301–320, 1987.

[3] P. Bjerksund and G. Stensland. Closed form valuation of American options. Working paper, NHH, 2002.

[4] O. Burkovska et al. Calibration to American options: numerical investigation of the de-Americanization method. *arXiv:1611.06181*.